

Exam Date & Time: 25-Jun-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Makeup Examination - June -2024

Design Patterns [CSE-4065]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) List the main design pattern categories and mention few of the patterns under each category. (2)
 - A)
 - B) Apply Model view controller architecture pattern in case of designing employee time sheet management software. (3)
 - C) Distinguish between structural and behavioural design patterns using any suitable example with respect to intent, problem, applicability, solution and meta UML model. (5)
- 2) List out the intent, problem, solution and meta UML model with suitable example for factory design pattern. (5)
 - A)
 - B) Apply factory design pattern and draw solution in the form of UML diagram for creating a logistics management application. (5)
- 3) List out the intent, problem, solution and meta UML model with suitable example for flyweight design pattern. (5)
 - A)
 - B) Apply Proxy design pattern and draw solution in the form of UML diagram for the scenario with YouTube manager to introduce lazy initialization and caching to a third party integration library which can handle multiple video downloads. (5)
- 4) List out the intent, problem, solution and meta UML model with suitable example for strategy design pattern. (5)
 - A)
 - B) Apply and draw meta model of Chain of responsibility design pattern with suitable example. (5)
- 5) Apply Prototype design pattern for the following scenario: (5)

Draw solution in the form of UML diagram for the scenario for building a drawing application capable of drawing different geometric shapes like circle, rectangle and

square.

A)

B)

Apply State design pattern for the following scenario:

Consider the state of a casual door in a factory and there is a single button to operate this door with below states:-

1. If closed, door begins opening
 2. If opening, another click begins closing
 3. Once open, 2 seconds later (timeout), the door begins closing and can be prevented by clicking after open state and before timeout begins
- (5)

Draw UML diagram by applying the state design pattern for modelling the event flow of different states such as door open, door closing, door stay open and door opening.

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